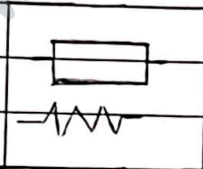


Unit 1: Electrical Safety.

Electric Circuit Symbols



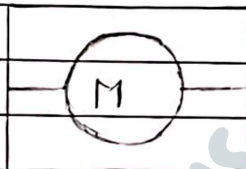
Resistor



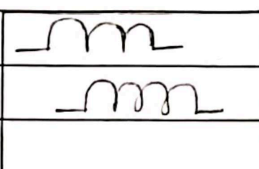
Ammeter



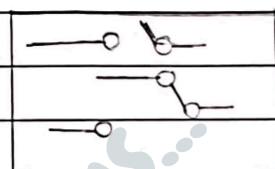
Voltmeter



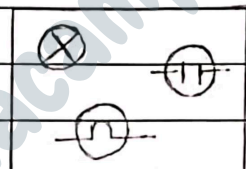
Motor



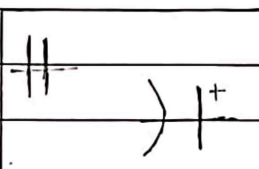
Inductor



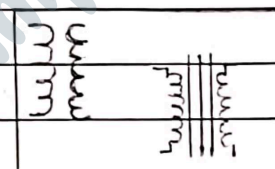
Switch



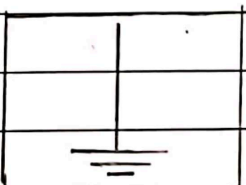
Lamp



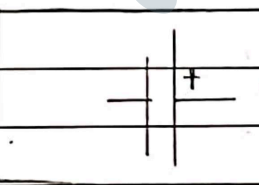
Capacitor



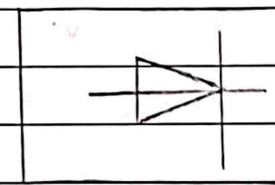
Transformer



Ground



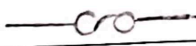
DC Voltage source

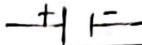


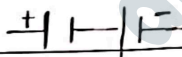
Diode

Electrical Circuit Symbols.

• Switch (open) → 

• Switch closed → 

• Cell → 

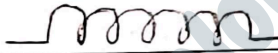
• Battery → 

• Diode → 

• Resistor → 

• Variable Resistor → 

• LED → 

• Inductor → 

• Lamp → 

• Fuse → 

• Voltmeter → 

• Ammeter → 

• Thermistor → 

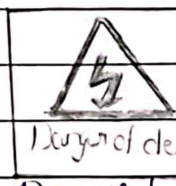
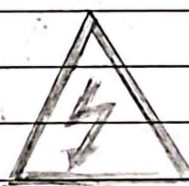
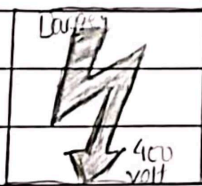
• Buzzer → 

Safety :- Safety is a state of being protected from potential harm or something that has been designed to protect and prevent harm. Safety equipment is designed to protect and prevent harm.

Safety Sign and Symbols

:- Electrical Safety Symbols are used to warn people that serious injuries can occur from burn, electric shocks and hazards. They also remind them to take the necessary steps to safe.

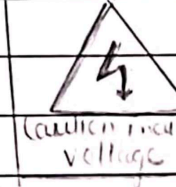
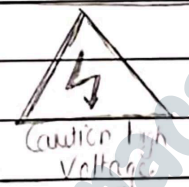
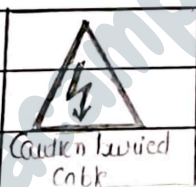
- Prohibition Sign :- Indicate certain behaviour is not allowed. (Red colour).
- Warning Signs :- Give warning about a specific hazard. ^{cyclone}
- Mandatory Action Signs :- Indicates that a particular action must be taken. (blue)
- Fire Safety Signs :- Provide information about fire safety.



Voltage Warning Labels

Electrical Voltage Symbol

Danger of death from electricity Warning



Buried Cable Warning

High Voltage Warning

Main Voltage Warning

7F To Study the use of Personal Protective Equipment (PPE)

To ensure electrical workmen from electrical hazards while working on Electrical equipment / System.
PPE is used! PPE is selected based on the voltage to which the employee may be exposed during work.

PPE types and their purpose

a) Hand protection: Rubber insulating gloves are used to prevent energy to the arm due to contact with live electric wires or burns due to electric arc.

b) Foot protection: Electrical Workers should wear leather footwear to protect their feet from electrical hazards.

c) Hearing protection: Hearing protective inserts are used to protect the ears of the employee from high sound levels of arc blast.

d) Face and Eye protection: Face shields with safety glasses or goggles must be worn to protect face and eye from flash injury.

e) Head protection: Non-conductive helmets should be worn whenever there is a danger of head injury from electric shock or burns.

Demonstrate How to free a person from electrocution and first aid for electric shock.

- Step 1 :-
- Turn off the power at the mains.
 - If this is not possible, stand on dry insulated material such as wooden stool, newspapers, books or rubber matting.
 - Push the casualty away from the source using non-conductive items.

Step 2:

Checking the casualty response if they respond by answering or moving, provided they are on no further danger, leave them in the position you found them.

Check for visible injuries and call for an ambulance.

Step 3:

If the casualty does not respond, open the airway by tilting their head back and lifting their chin. Check the air way and remove any obvious obstruction from his/her mouth.

Step 4 :

Check for signs of breathing by looking for the chest movements. Listen at the mouth for breathing sounds and feel for air on your cheek. Look, listen and feel for 5 seconds.

Step 5:

Feel the pulse for 5 second if the pulse and breathing are present, place in the recovery position. If pulse is present and breathing is absent start rescue breaths.

- If pulse and breathing are absent, Commence CPR (Cardiopulmonary Resuscitation), Whilst waiting for the ambulance, Give 30 chest compression and 2 breaths. Continue this after 2 breath time and continue till ambulance arrives or the casualty recovers.

7) To free a person from electrocution in first aid for electric shock follows these steps:-

Step 1:

Ensure Safety :- Before approaching the victim, make sure such they are no longer in contact with the electrical source, if possible switch off the power.

Step 2:

Use insulating objects :- If you can't turn off the power, wear rubber gloves and boots or stand on an insulating surface before touching the victim. Use non-metal object like a wooden pole to remove any entangled electrical lines.

Step 3:

Check Vital signs :- Immediately check for breathing and heart beat. If the victim is not breathing, perform artificial respiration. If there's no heartbeat, apply chest compression.

Step 4

Cool burns :- If the victim has burns, cool the affected areas with cold water and cover them with a clean, dry, non fluffy dressing.

Remember electrical shocks can be serious, so seek medical assistance if the victim shows any concerning symptoms or signs of injury.

Administer appropriate first aid to victims, bandaging, heart attack CPR etc.

Date :- 29/04/2024

Unit - II

PAGE NO.	
DATE	

Fundamental of EL & EC Engineering Unit - II (Electrical Fundamental)

Topic :- Electricity

- Electric charge :- is the property of sub-atomic particles that causes it to experience a force when placed in an electromagnetic field.

There are two type of charge :-

- * Positive charge
 - * Negative charge
- (Same charge repel each other and different charged attract each other).

- * S.I unit of charge is Coulomb (C).
- * 1 Coulomb means 6.24×10^{18} electron.
- * An electron has ~~highest~~ negative charge of $1.6 \times 10^{-19} \text{C}$.

✓ Electric Current :- The rate of flow of electric charge through a conductor.

It is denoted by I or i .

$$\text{i.e. } I = \frac{dQ}{dt} \quad \text{or} \quad I = \frac{Q}{t}$$

Its S.I unit is ampere (A). [Ammeter].

✓ Electric potential :- Electric potential at a point within an electric field is the amount of work done in bringing a unit +ve charge from infinite to the point.

It is denoted by V .

Its S.I unit is volt (V). [Measured by Voltmeter]

Resistance :- • Voltmeter is always connected in parallel.
It is opposition or the obstruction offered by conductor to the passage of drifting electron.

ORGE NO	
DATE	

✓ Potential difference :- Potential difference between two point within an electric field in the amount of work done in bringing a unit tri charge from one point to another point.

✓ Resistance of a material is its property by virtue of which it opposes the flow of current through it.

It is denoted by R or Ω .

Its S.I unit is ohm (Ω).

Measured by ohmmeter. (Megger - Surface).

✓ Ohm's law :- The relation between potential difference and current is called Ohm's law.

Potential difference $\propto$ Electric Current.

$$\begin{aligned} V &\propto I \\ [V &= I \cdot R] \end{aligned}$$

✓ Voltage :- The "pressure" that pushes electricity.

✓ EMF (electromotive force) :- The total difference in electrical potential between the terminal of cell, when it is not doing any external work.

✓ [When no current is drawn from a cell, the potential difference between the terminal of the cell is called its emf.]

• Describe the source of electrical energy.

i) Hydro power

⇒ Hydro power is electricity generated by harnessing the energy of flowing or falling water. This energy is converted into electrical power using turbines and generators.

ii) Thermal power plant

⇒ A thermal power plant is a place where electricity is made using heat. They burn things like coal or gas to create steam which turns to generate electricity.

iii) Nuclear power

⇒ Nuclear power is electricity made by splitting atoms. This process releases a lot of heat, which is used to make steam and turn turbines for generating electricity.

iv) Petroleum & gas power plant

⇒ A petroleum and gas power plant is a place where electricity is produced by burning oil or natural gas. This burning creates heat which is used to make steam and turn turbines to generate electricity.

v) ~~Wind power~~ ^{Solar power} plant

⇒ A ~~wind~~ ^{solar} power plant is a place where electricity is made using sunlight panels called solar panels. Soak up sunlight and turn it into electricity.

vii) Wind power plant

⇒ A wind power plant is a place where electricity is made using wind. Big wind mills catch the wind and spin, turning turbines to generate electricity.

viii) Bio-gas plant

⇒ A biogas plant is where gas is made from waste. It used things like food scraps, manure or plants which break down and produce gas that can be used for cooking or making electricity.

viii) Geo-thermal energy

⇒ Geothermal energy is heat from the earth. We use it to make electricity or to heat buildings.

• Electrical Work

3) • Mention meters used to measure different electrical quantities:

⇒* Ammeter, Voltmeter, Wattmeter, Ohmmeter, Digital multimeter, Megger, Tong tester.

4) Explain Supply System like AC and DC.

⇒ AC (Alternating Current)

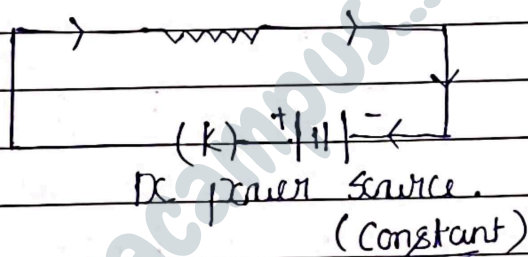
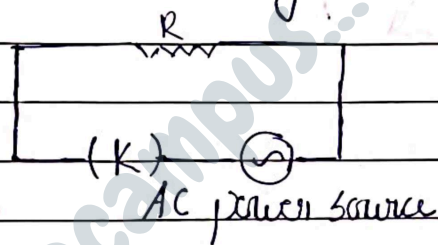
- AC is easy to transfer over longer distance even between two cities without much energy loss.
- The rotating magnets cause the change in direction of electric flow.
- The frequency of AC is dependent upon the

Country, But generally the frequency is 50 Hz or 60 Hz.

- In AC the flow of current changes its direction forward and backward periodically.

DC (Direct Current)

- DC cannot be transferred over a very long distance. It loses electric power.
- The steady magnetism make DC flow in a single direction.
- DC has no frequency or Zero frequency.
- It flows in single direction.



Relationship between V, I and R.

- Ohm's law states that current flowing through the conductor is directly proportional to voltage across the conductor. It is inversely proportional to the resistance of the conductor, kept temperatures remain constant.

$$I = \frac{V}{R}, \quad V = IR, \quad R = \frac{V}{I}$$

Describe open circuit, close circuit and short circuit:

- Open circuit :- An open circuit is a circuit in which the path is incomplete which means there is no flow of current.

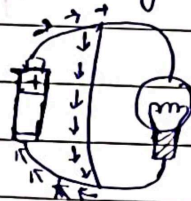


$$I \propto \frac{1}{R}$$

- Closed Circuit : A closed circuit is a circuit in which the path is complete and there is flow of current.



- Short Circuit : A short circuit is a circuit which offers very little resistance to current flow and can cause dangerously high current flow through a circuit.



Electric Circuit :- A continuous conducting path, between the terminals of source of electricity such as cell, battery is called an electric circuit.

Behaviour of V, I in series and parallel DC circuit.

• Series DC Circuit

- ⇒ ~~Like~~ In a series DC circuit, the same current flows through all components and the total voltage is the sum of individual voltage drops across resistor.
- The voltage is divided among resistors, with different voltage drops across each based on their resistance value.
- The total effective resistance in a series circuit is the sum of individual resistance.
- The behaviour of voltage and current in a series DC circuit follows ohm's law where ~~Vot~~ Voltage

is directly proportional to resistance and current is constant throughout the circuit.

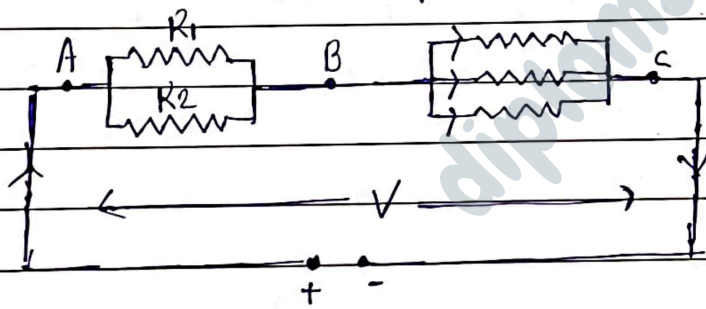
• Parallel DC Circuit :

⇒ In a parallel DC circuit, components share a common voltage, but have different currents flowing through them. The total current drawn from the source equals the sum of currents through each component.

• The voltage across all components in parallel is the same as the source voltage.

• Current in parallel circuit divides among the paths available, flowing through each inversely proportional to resistance.

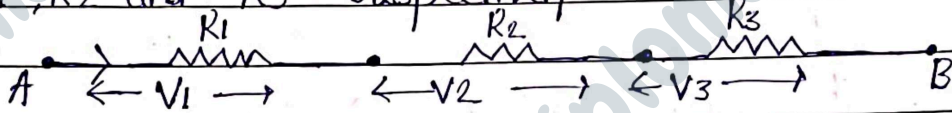
• The total circuit current equals the sum of individual branch current in a parallel circuit.



DC Series, parallel circuit.

Equation to find the effective Resistance connected in Series :

⇒ Consider three resistor in series as shown, with resistance R_1 , R_2 and R_3 respectively



If V_1 , V_2 and V_3 are potential difference across them respectively, the total potential difference across the system of resistor.

$$V = V_1 + V_2 + V_3$$

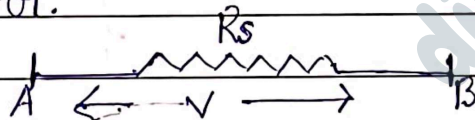
~~(Since the current through them in the same)~~.

$$\text{i.e. } V = IR_1 + IR_2 + IR_3$$

$$= I(R_1 + R_2 + R_3) \quad \text{--- (1)}$$

(Since the current through them in the same).

Let the three resistors be placed by a single resistor of resistance R_s called the equivalent or effective resistor.

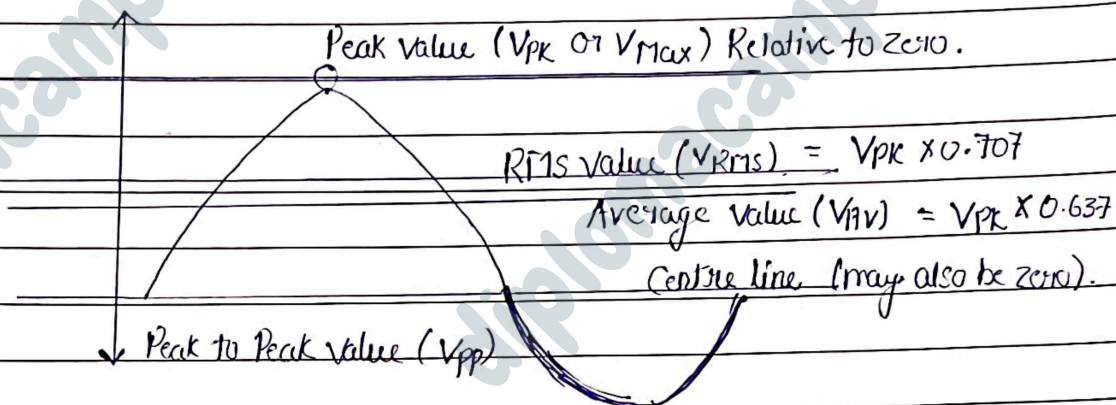


$$\text{Hence for equivalent resistor, } V = IR_s \rightarrow (2)$$

from (1) and (2).

$$\text{We have, } R_s = R_1 + R_2 + R_3.$$

AC Sine Wave



Periodic time T (frequency = $\frac{1}{T}$)

Sinusoidal Voltage

Alternators produce sinusoidal alternating voltage i.e. alternating voltage is a sine wave.

A sinusoidal alternating voltage can be produced by rotating a coil with a constant angular velocity in a uniform magnetic field.

The sinusoidal alternating voltage can be expressed by the equation,

$$V = V_m \sin \omega t$$

Where

V = Instantaneous value of alternating voltage

V_m = Max. value of alternating voltage

(omega) ω = Angular velocity of the coil. ($2\pi n$)

Sinusoidal voltage always produces sinusoidal currents, unless the circuit is non-linear.

Therefore, a sinusoidal voltage can be expressed in the same way as voltage i.e. $i = I_m \sin \omega t$.

i = Instantaneous Current I_0 = peak value of " " i = Instantaneous Emf E_0 = Peak value of " "
--

Resistor $V = \frac{I}{\sqrt{2}}$
Inductor $\frac{1}{\sqrt{2}}$

(1.5 volt) $\rightarrow$ Cell

PAGE NO.	
DATE	

Amplitude / Peak Value / Maximum Value

Amplitude of an Alternating Current (voltage or current) is the instantaneous value of that alternating quantity when it reaches its maximum value, either in the positive or negative direction. It occurs twice during each cycle.

• Time period :- Time taken by an alternating current to complete one cycle is called the time period.
($T = 1/f$).

• Frequency :- The number of cycles that a sine wave complete in one second is measured in Hertz (Hz).

(Q31) Phase / Phase angle :- The development of an alternating quantity (current or voltage) through different stage is known as phase or phase angle.

• Phase difference : The phase difference refers to the time intervals by which one wave is behind or ahead of the waveform.
Expressed in electrical degree or radians.

Electrical Work :- When certain charge is transferred from one point to another point, in an electric circuit, against a potential difference, the electrical work is said to be done or electrical energy expended.

Unit :- Joule / Watt Second

In other word :- If a charge of Q Coulombs is transferred between two points of an electrical circuit against a potential difference of E -Volts, then electrical work is done.

∴ It is measured by using Energy meter in kWh.

Electric power :- The rate of doing electric work is called electric power.

$$\text{Electric power} = \frac{\text{Electric Work done}}{\text{Time taken.}}$$

S.I unit of electric power is watt.

When one joule of energy is used for 1 second, electric power is equal to one watt.